Mitchell Gray

meg346@cornell.edu mgray88@gatech.edu MitchellEGray MitchellGray100

Education

Georgia Tech, Atlanta, GA — **Jan 2025 – Aug 2026 (Part-Time)**
Master's of Science, Computer Science — *GPA: 4.0; Artificial Intelligence*

Cornell University, Ithaca, NY — **Aug 2020 – May 2024**
Bachelor's of Science, Computer Science — *Distributed Systems*
Minor, Operations Research Information Engineering — *30 Extra Graduate-Level Credits*

Knowledge Domain: Distributed Systems · Systems Programming · AI · Machine Learning · Databases · Cloud

Technical Skills

Languages: C · C++ · Python · Java · Rust · PL/SQL
Software & Tools: Git · Pytorch · Scikit-learn · NumPy · Pandas · Poetry · CMake · CI/CD · Docker · K8s · Azure · AWS

Relevant Work Experience

Oracle — **Redwood City, CA**
Senior Software Engineer | **C, PL/SQL, Raft, Java, OracleDB, OCI** — **Sept 2026 – Present**

- Working on Raft Replication in the Oracle Globally Distributed Database for Shard Replication · Writing fault-tolerant and scalable Distributed Systems code in **C**, **Java**, and **PL/SQL**.
- placeholder placeholder placeholder placeholder placeholder placeholder placeholder placeholder placeholder placeholder placeholder placeholder · placeholder placeholder placeholder placeholder placeholder placeholder.

Software Engineer II — **Sept 2025 – Sept 2026**

- Designed and implemented a generic shard-operations replication framework that consolidated multiple ad-hoc code paths into a single framework · Improved maintainability, scalability, and ease of creating new operations with fewer defects.
- Received **Training & Mentorship Excellence Award** · Provided technical mentorship to our newest engineer.
- placeholder placeholder placeholder placeholder placeholder placeholder placeholder placeholder placeholder placeholder placeholder placeholder placeholder · placeholder placeholder placeholder placeholder placeholder placeholder.
- Fixed critical vulnerabilities in slow-follower throttling functionality · Implemented new throttling algorithms.

Software Engineer I — **June 2024 – Sept 2025**

- Engineered a fault-tolerant replica replication feature on Oracle Cloud Infrastructure · Enforced placement of Raft replicas on separate server racks to eliminate single-rack replication-unit failures and improve database availability.
- Refactored over 1,200 lines of Sharding serialization code to eliminate a critical data-loss defect · Designed robust de/serialization pipelines and multi-format conversion logic that greatly improved data integrity and system reliability.
- Optimized multiple replica-to-replica data transfer operations · Achieved a **700%** performance improvement without workload and **250%** improvement under active workloads after optimization code changes.
- Resolved critical recoverability issue in replica-to-replica data transfer operations by ensuring complete transfer of essential recovery-related tables and data between replicas · Improved reliability and system integrity during failure.
- Implemented comprehensive Pluggable DBs lifecycle management · Enabled seamless Drop, Unplug, and Plug operations of PDBs to streamline maintenance for Raft replicas using Oracle's multi-tenant container architecture.
- Enhanced error-handling mechanisms throughout the codebase · Improved error handling reliability and diagnosability.

Gecko Robotics — **Pittsburgh, PA**
Software Engineer Intern | **C++, Python, C, CMake, GCP, CI/CD** — **May 2023 – Aug 2023**

- Worked on the Robot Controls team · Wrote code for an asynchronous distributed system in **C++**, **Python**, and **C**.
- Revamped Robot & DAQ emulators · Implemented communications protocol and Client/Server TCP networking.
- Replaced VS build-system with **CMake** · Added calibration support to emulators · Integrated **CI/CD** and **Poetry**.

Carnegie Mellon University - Software Engineering Institute — **Pittsburgh, PA**
DevOps Engineer Intern | **Python, Rust, CI/CD, Docker, Kubernetes** — **May 2022 – May 2023**

- Developed **Python**, **Rust**, and **Bash** code on various NDA projects · Updated and created **CI/CD** pipelines · Created GitLab **REST API** live data visualizations for clients · Used ArgoCD and Helm to deploy AWS EKS cluster.
- Built internal analytics tools using **Python**, **Neo4j**, NeoDash, and PageRank-based graph analysis to generate engineering metrics and documentation adopted across multiple teams within the company.

Relevant Academic Work Experience

Georgia Tech Contextual Computing Group — **Atlanta, GA**
Graduate Researcher | **AI, ML, Python, Typescript** — **June 2025 – Aug 2026**

- Worked on Nosi, an IDE that combines an encrypted virtual file system and application behavior-logging · Utilizes **Machine Learning** techniques to analyze behavioral data and flag potentially-cheating students.
- Published at **Learning at Scale 2026** · Integrated the IDE into multiple Georgia Tech graduate-level CS courses.

Georgia Tech — **Atlanta, GA**
Graduate Teaching Assistant | **AI, Python, NumPy** — **May 2025 – Aug 2026**

- Taught graduate-level **Artificial Intelligence** · Helped with new plagiarism-detection tooling integration.

Cornell University — **Ithaca, NY**
Undergraduate Teaching Assistant | **Python, NumPy, SQL, Java** — **Aug 2022 – May 2024**

- Received the **Course Staff Exceptional Service Award** · Taught Databases, Object-Oriented Programming, and Physics · Lectured 400+ students with custom-designed lecture content · Led Exam sessions and other TAs.

Cornell Database Group — **Ithaca, NY**
Undergraduate Researcher | **ML, Java, AWS** — **Jan 2022 – May 2024**

- Published and presented paper at **VLDB 2023** combining **Reinforcement Learning** and Worst-Case Optimal Joins.
- Worked on upgrading query engine to a **Distributed Architecture** · Created dynamic statistics & RL visualizations.

Publications

Learning at Scale (2026) — [https://dl.acm.org/doi/abs/10.14778/3611540.3611629]
Nosi IDE: Securing Coding Assessment Integrity in the Age of LLMs via Process-Driven Analytics

VLDB (2023) — [https://dl.acm.org/doi/abs/10.14778/3611540.3611629]
Demonstrating ADOPT: Adaptively Optimizing Attribute Orders for Worst-Case Optimal Joins via Reinforcement Learning

Awards

Oracle, Database Organization — *Mentorship and Leadership Excellence Award*
Recognizes outstanding efforts in mentorship and training of others that go above and beyond the call of duty. Multi-day, multi-recipient efforts beyond the direct organization.

Cornell Bowers CIS — *Course Staff Exceptional Service Award*
Faculty-nominated award for exceptional commitment, hard-work, dedication, and exceptional service of a Teaching Assistant during the 2023-2024 academic year.

Relevant Projects

New AI Infra Project Placeholder | Personal Project | ***AI, Go*** | *Github* — **May 2026 – Present**

- 1 of 44 projects selected for BOOM 2024. My team and I created a Scalable Platform for Efficient Execution of Distributed testing. The fault-tolerant system contains a controller node that orchestrates worker nodes that run **JUnit** tests on **Java** code. The worker nodes report their findings to the controller. Once all tests are ran, test results are shown in the frontend to the user.

Improving 3D Point-Cloud Classification | Deep Learning | ***AI, Python, Pytorch*** | *Github* — **Aug 2025 – Dec 2025**

- For this project, we re-implemented and upgraded state-of-the-art point cloud classification Deep Neural Networks. We started by implementing PointNet and PointPillars. Then we upgraded them by modifying their architecture, adding Dropout, implementing Data augmentation, and introducing Residual Connections. We improved validation accuracy by 4% and massively reduced overfitting.

Distributed Testing Platform (SPEED) | Masters Project | ***Java*** | *Github* | *BOOM* — **Aug 2023 – May 2024**

- 1 of 44 projects selected for BOOM 2024. My team and I created a Scalable Platform for Efficient Execution of Distributed testing. The fault-tolerant system contains a controller node that orchestrates worker nodes that run **JUnit** tests on **Java** code. The worker nodes report their findings to the controller. Once all tests are ran, test results are shown in the frontend to the user.

SpGEMM Parallelization on GPUs | Parallel Computing | ***CUDA, C++*** | *Github* — **Jan 2024 – May 2024**

- In this project, our team parallelized sparse-matrix multiplication operations using GPUs and experimented with multiple variations of the algorithm, massively improving performance of the algorithms.